1.1 Measurable spaces

Sample Space

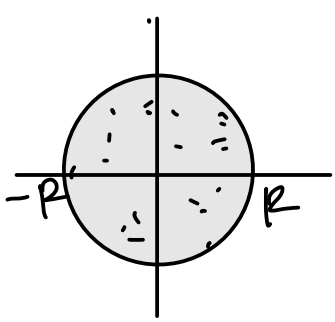
The set of all possible outcomes of any "experiment" is called sample space Ω. The sample space can be discrete or continuous.

Example Discrete Sample space

- Flipping a coin: $\Omega = \{H, T\}$
- Rolling a die: $\Omega = \{1, 2, 3, 4, 5, 6\}$
- Flipping two coins: $\Omega = \{HH, HT, TH, TT\}$
- How many times we need to flip two coins to get HH? $\Omega = \{1, 2, 3, \dots\}$

Example Continuous Sample space

Throwing darts



$$\Omega = \{(x,y) \in \mathbb{R}^2 : x^2 + y^2 = R^2\}$$

uncountable set

There are infinite points inside this disk.

Events

*An **event** is any subset of a sample space.*

Example Rolling a die: $\Omega = \{1,2,3,4,5,6\}$

- Event 1: $A = \{\text{even outcomes}\} = \{2,4,6\}$
- Event 2: $B = \{\text{odd outcomes}\} = \{1,3,5\}$
- Event 3: $C = \{\text{outcomes greater than } 4\} = \{5,6\}$

Say we roll the die and get

$3 \rightarrow$ Event 2

$2 \rightarrow$ Event 1

$5 \rightarrow$ Event 2 <u>and</u> event 3

Power Set

The collection of all possible events (subsets) of Ω, including the empty set $\emptyset$ and Ω itself, is called power set, $\mathcal{P}(\Omega)$.

Example: If $\Omega = \{a,b,c\}$ then

$$\mathcal{P}(\Omega) = \{\emptyset, \Omega, \{a\}, \{b\}, \{c\}, \{a,b\}, \{a,c\}, \{b,c\}\}$$

$\sigma-$algebra

Let $I = \{1, 2, 3, ...\}$ be a countable set of indices. A $\sigma-$ algebra, $\mathcal{F}$, is a collection of events of Ω with the following properties:

(i) $\Omega \in \mathcal{F}$,

(ii) If $A \in \mathcal{F}$, *then* $A^c \in \mathcal{F}$, (closed under complementation)

(iii) If $A_i \in \mathcal{F}$, *then* $\bigcup_{i \in I} A_i \in \mathcal{F}$. (closed under countable unions)

Example Let $\Omega = \{1, 3, 5\}$. Is $\mathcal{F} = \{\emptyset, \Omega\}$ a σ-algebra?

Check if (i), (ii) and (iii) are satisfied:

(i) $\Omega \in \mathcal{F}$ ✓

(ii) $\Omega \in \mathcal{F} \Rightarrow \Omega^c = \emptyset \in \mathcal{F}$ ✓

$\emptyset \in \mathcal{F} \Rightarrow \emptyset^c = \Omega \in \mathcal{F}$ ✓

(iii) $\Omega \cup \emptyset = \Omega \in \mathcal{F}$ ✓

$\mathcal{F}$ is a σ-algebra

Example Let $\Omega = \{a, b, c\}$. $\mathcal{P}(\Omega)$ is also a σ-algebra (prove it)

Theorems

- $\emptyset \in \mathcal{F}$
- $\mathcal{P}(\Omega)$ is a σ-algebra
- If A_n, $n \in I \Longrightarrow \bigcap_{n \in I} A_n \in \mathcal{F}$ closed under countable intersections
- The intersection of σ-algebras is a σ-algebra.

Remarks

- A σ-algebra is a collection of events that are available and we are interested in.
  In other words, a σ-algebra contains all the interesting information the experimet provides

- This collection of events, this algebra, <u>must</u> make sense.
  - Contains $\emptyset$ and Ω
  - Basic operations:

    "not" A, A^c (negation)

    "or" Union of sets (disjunction)

    "and" Intersection of sets (conjuction)

  - It is closed under these operations

Measurable Space

Let Ω be a nonempty set and $\mathcal{F}$ a $\sigma-$ algebra on Ω. The pair $(\Omega, \mathcal{F})$ is called measurable space.

An $A \in \mathcal{F}$ is called $\mathcal{F}$-measurable set

Example

Let $\Omega = \{a, b, c\}$. We showed that $\mathcal{P}(\Omega)$ is a σ-algebra on Ω. Thus $(\Omega, \mathcal{P}(\Omega))$ is a measurable space